

Vikas Pathak

Email: vikaspathak0911@gmail.com Ph: +91 9170135960 GitHub: [pathakx](#) LinkedIn: [vikaspathak11](#) Portfolio: [vikaspathak11.github.io](#)

B.Tech CSE graduate from VIT (CGPA: 8.68) with hands-on experience in building AI automation systems and Retrieval-Augmented Generation (RAG) applications using Python, LangChain, LangGraph, React, FastAPI, Flask, Pinecone, and AWS. Built projects like an autonomous Quality Analyst agent (test case + Selenium script generation), medical chatbot, and EduMorph — RAG-based AI Learning Platform. Strong in DSA, SQL, OOP with 300+ LeetCode problems solved.

TECHNICAL SKILLS

Languages: Python, Java

Frameworks: Flask, FastAPI, LangChain, LangGraph

Skills: Machine Learning, Agentic AI, RAG, SQL, DSA, OOPs

Certifications: OCI Generative AI Professional Certificate (2025), IBM Cyber Security Analyst (2025), Cloud Computing (NPTEL), Marketing Analytics (NPTEL), Japanese Language Proficiency Test N5 Level

EXPERIENCE

Tsuroni Ltd. — Israel — *AI Workflow Automation Intern* — Remote

April 2025 – June 2025

- Architected multi-agent AI workflows on n8n, built SEO/content automation pipelines, automated competitor scraping + LLM blog generation, invoice extraction to Airtable using Gemini, and stock-analysis workflows integrating technical + sentiment analysis, reducing manual work by 70%, increasing content output by 300%, and improving insight reliability by 35%.

EDUCATION

VIT Bhopal University

2022 – 2026

Bachelor of Technology in Computer Science & Engineering (CGPA: 8.68)

—*Bhopal, MP*

PROJECTS

EduMorph — RAG-based AI Learning Platform — FastAPI, React, ChromaDB, LLM

Feb 2026

- Built a production-ready RAG system using semantic search over user-uploaded documents, improving answer relevance and contextual accuracy.
- Reduced LLM hallucinations by grounding responses with top-k retrieved chunks and source attribution.
- Designed a scalable vector retrieval pipeline using MiniLM embeddings and ChromaDB for efficient document indexing and querying.
- Developed an end-to-end learning platform (QA, flashcards, quizzes) with FastAPI backend and React frontend, enabling structured knowledge retrieval and active recall.
- Link: <https://github.com/pathakx/EduMorph>

Quality Analyst AI Agent — FastAPI, React, Tailwind, LangGraph, RAG, Pinecone, LLM

Aug 2025

- Built an autonomous QA agent that generates end-to-end test cases and Selenium Python scripts from requirements (PDF/MD/TXT) and HTML UI files using RAG and Langgraph.
- Reduced manual testing effort by automating test design and script generation via LLM pipelines (Gemini/Groq) with context-aware retrieval.
- Designed a unified FastAPI backend serving both API and React + Tailwind frontend, enabling seamless document upload, test generation, and script export.
- Implemented a knowledge-grounded testing workflow that converts product documentation into executable automation scripts, improving test coverage and consistency.
- Link: <https://github.com/pathakx/QA-Agent>

Medical ChatBot — RAG, LLMs, LangChain

Feb 2025

- Built a Medical Chatbot using LangChain + Flask with LLM-based QA over domain documents using RAG.
- Implemented vector embeddings storage + semantic retrieval with Pinecone to improve response grounding.
- Designed an end-to-end Dockerized deployment pipeline using AWS ECR + EC2, enabling scalable production hosting.
- Automated deployment with AWS CI/CD using GitHub Actions + self-hosted EC2 runner, managing secrets securely via GitHub Secrets.
- Link: <https://github.com/pathakx/medical-chatbot>

ACHIEVEMENTS

- Qualified Meta Hacker Cup Season 2025 Round 3 (Rank-148).
- Qualified TCS CodeVita Season 13 (Round 1), 2025 & TCS HackQuest Season 10 (Round 2), 2025.
- Selected for Student Internship Program, Bhopal Police (2024-2025).